

1(a)(i)	C-N bond shown ; =O shown on C and -H shown on N of peptide group ; water shown as a product ;	3
1(a)(ii)	peptide ; A amide R polypeptide bond	1
1(a)(iii)	ribosome ; I 70S / 80S A <u>rough endoplasmic reticulum</u> I rough ER / RER	1
1(b)(i)	any two from: alpha- / α -, helix, helices ; R A or a <i>for alpha</i> beta- / β -, pleated sheet(s) / pleat(s) ; A beta / β , sheet(s) R B or b <i>for beta</i> region of no fixed shape / AW ; A random coil	2

Question	Answer	Marks
1(b)(ii)	any two from: 1 protein is, a single <u>polypeptide</u> / made of one chain of amino acids or quaternary structure is two or more <u>polypeptides</u> / more than one <u>polypeptide</u> ; 2 interactions / bonds, between, R-groups / side chains, (of amino acids) on the same <u>polypeptide</u> or 3 quaternary has, interactions / bonds, between, R-groups / side chains, (of amino acids) on <u>different polypeptides</u> ; 4 AVP ; e.g. no prosthetic group	2
1(b)(iii)	<i>secondary structure</i> 1 in alpha-helix each N-H forms hydrogen bond with C=O three or four amino acids apart in same polypeptide ; 2 in beta-pleated sheet each N-H forms hydrogen bond with C=O on adjacent part of the polypeptide ; <i>if mp1 or mp2 not awarded</i> 3 (by) hydrogen bonds (in secondary structure) ; <i>tertiary structure</i> 4 hydrogen bonds between, R-groups with C=O and HN groups / carboxyl and amine groups ; A between polar R groups 5 ionic bonds between R-groups with NH_3^+ and COO^- groups ; A between charged R groups / AW 6 hydrophobic interactions between non-polar R-groups ; <i>if mp4, mp5 or mp6 not awarded</i> 7 (named) interactions / bonds, between R-groups (in tertiary structure) ;	3